Celestial Mechanics

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

8 lecture sections · 15 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; 15 graded problems and independent worked solutions turn the ideas into practice.

▼ **Contents & learning path**

1. Gravity: a force that points somewhere
2. Both bodies move: the barycentre
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1. Gravity: a force that points somewhere

A planet does not keep moving in a straight line around the Sun because something pushes it forward. It keeps falling toward the Sun while its sideways motion carries it past the Sun. Gravity supplies the inward acceleration; inertia supplies the continued motion.

For two point masses, or outside spherically symmetric bodies,

$$F = \frac{Gm_1m_2}{r^2}, \qquad g(r) = \frac{GM}{r^2}. \tag{1}$$

The distance r is measured between centres, not between surfaces. The force is equal and opposite on the two bodies. A small acceleration can still describe a large orbit if it acts for a long time.

Constants and useful shortcuts. Use $G = 6.6743 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $M_\odot = 1.98847 \times 10^{30}\ \mathrm{kg}$, $1\ \mathrm{AU} = 1.495978707 \times 10^{11}\ \mathrm{m}$, and one Julian year $= 365.25$ days. For solar orbits, $T^2 = a^3/M$ when T is in years, a in AU and the **total** mass M in solar masses. The circular speed at 1 AU around one solar mass is $29.7847\ \mathrm{km\,s^{-1}}$.

2. Both bodies move: the barycentre

The centre of mass is the balance point of the system. For two masses separated by r,

$$m_1 r_1 = m_2 r_2, \qquad r_1 + r_2 = r. \tag{2}$$

The heavier body has the smaller orbit. In the centre-of-mass frame the relative separation obeys a one-body gravitational problem with $G(m_1 + m_2)$. This is why a binary's period measures the **sum** of its masses. A star's small wobble can reveal an unseen companion.

3. Circular motion and escape

For a circular orbit, gravity is exactly the required centripetal acceleration:

$$v_c = \sqrt{\frac{GM}{r}}, \qquad T = 2\pi\sqrt{\frac{r^3}{GM}}. \tag{3}$$

The orbital radius is the distance from the centre. Escape is an energy condition, not a requirement to keep firing an engine. With zero potential energy at infinity,

$$\epsilon = \frac{v^2}{2} - \frac{GM}{r}, \qquad v_{\mathrm{esc}} = \sqrt{\frac{2GM}{r}} = \sqrt{2}\, v_c. \tag{4}$$

Negative ϵ means bound; zero is the parabolic escape limit; positive means a nonzero speed at infinity. Here ϵ is energy per unit mass in the one-body approximation.

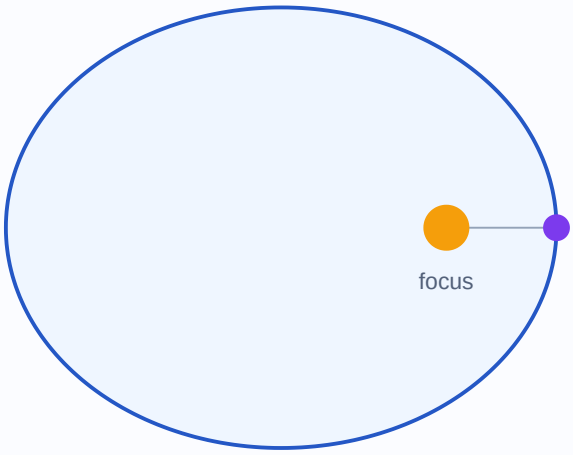
INTERACTIVE EXPERIMENT

Change the orbit, not the conservation laws.

Relative two-body orbit in AU, solar masses and years. The primary is placed at a focus; the plotted path is the separation vector’s orbit. Sizes are symbolic.

Period **1.000 yr**	Periapsis **0.400 AU**	Apoapsis **1.600 AU**
Periapsis speed **59.57 km/s**	Apoapsis speed **14.89 km/s**	Specific relative energy **-443.6 km²/s²**

Two-body relative orbit · animation slowed



$r_p = a(1 - e)$ $r_a = a(1 + e)$ $T^2 = a^3/M$ (AU, yr, $M_\odot$)

At fixed total mass and semimajor axis, eccentricity changes the shape and speed range but not the period. Current period: 1.000 years.

Try this. Double the semimajor axis at fixed mass. By what factor does the period change? At fixed semimajor axis, increase eccentricity: does the period change?

Snapshot of the experiment. Open the HTML edition to change the controls.

4. Ellipses: the same orbit, two conservation laws

An ellipse has semimajor axis a, eccentricity e, periapsis $a(1-e)$ and apoapsis $a(1+e)$. The central mass is at a focus. Kepler’s equal-area law is angular-momentum conservation: the body moves faster when closer.

$$T^2 = \frac{4\pi^2 a^3}{G(m_1+m_2)}, \qquad v^2 = G(m_1+m_2)\left(\frac{2}{r} - \frac{1}{a}\right). \tag{5}$$

The second expression is the **vis-viva equation**. At $r = a$, the speed is $\sqrt{GM/a}$ whatever the eccentricity. The direction is different, but the energy is fixed by a. At the two apsides the velocity is perpendicular to the radius, so $r_p v_p = r_a v_a$.

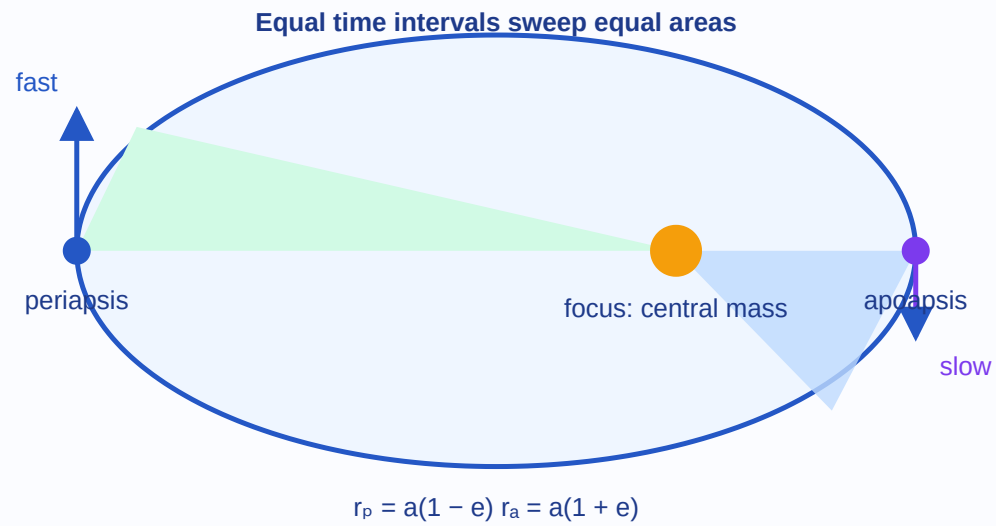


Figure 1. The force points toward a focus, not toward the centre of the ellipse. The two shaded sectors illustrate equal areas in equal times; the ellipse is schematic.

5. Changing an orbit costs energy

A short tangential burn changes the speed at one radius, and therefore the opposite side of the orbit. A Hohmann transfer between circular coplanar orbits uses half an ellipse tangent to both. If the radii are $r_1 < r_2$, its semimajor axis is $(r_1 + r_2)/2$. Use vis-viva to find the transfer speeds, then subtract the local circular speeds. The transfer time is half the transfer period. This idealization ignores launch, planetary gravity, inclination changes and timing.

6. Tides and the Roche limit

Tides arise because gravity is not identical across an extended body. For a small radius R at distance d, the centre-to-near-side differential acceleration is approximately

$$\Delta g \simeq \frac{2GMR}{d^3}. \tag{6}$$

The near-to-far difference across the whole diameter is twice this value, $4GMR/d^3$. The inverse cube makes tides much more sensitive to distance than the force itself. The Roche limit estimates where tidal stretching overwhelms a satellite's self-gravity. For an ideal fluid satellite,

$$d_{\mathrm{Roche}} \simeq 2.44 R_M \left(\frac{\rho_M}{\rho_m}\right)^{1/3}. \tag{7}$$

A simple rigid, strengthless estimate uses a coefficient near 1.26. Real material strength, spin, shape and orbital history matter. The formula is a model-dependent scale, not a universal destruction switch.

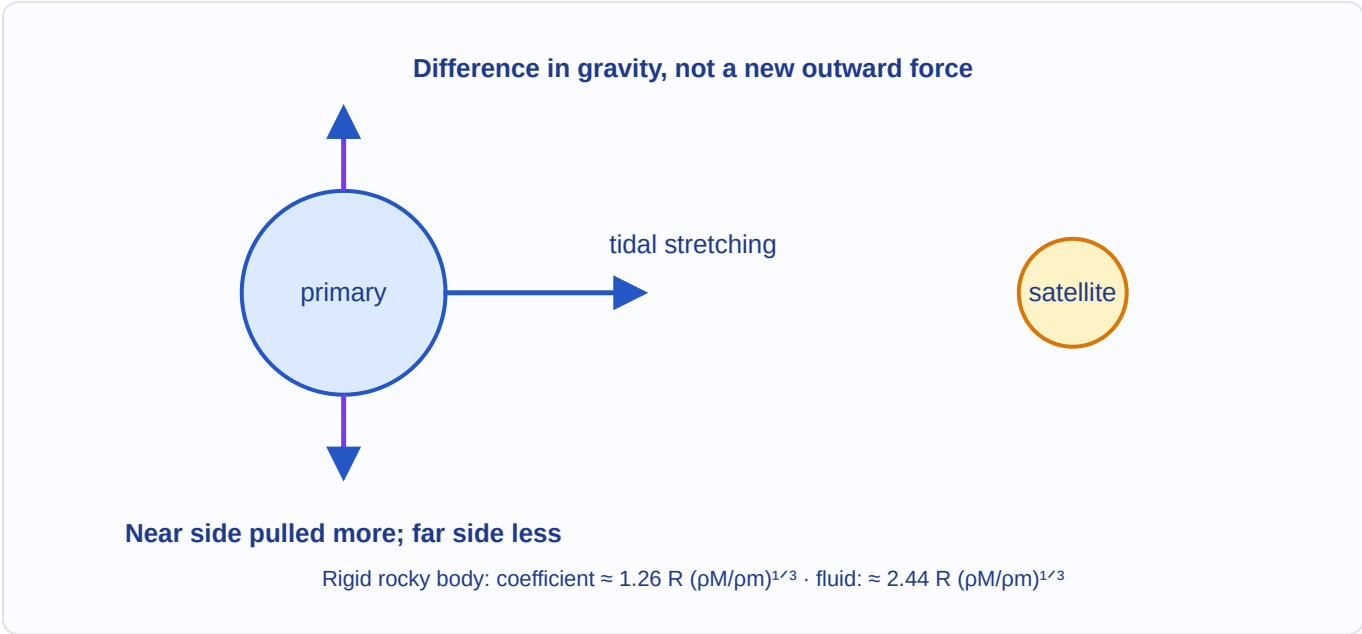


Figure 2. The Roche limit is a scale, not a universal sharp boundary. Material strength, spin and orbit matter; the quoted coefficients are idealized models.

7. Lagrange points: balance in a rotating frame

In a frame rotating with a circular binary, five stationary points exist. L1, L2 and L3 lie on the line through the masses; L4 and L5 form equilateral triangles with them. These are not places where gravity vanishes. Gravity and the rotating-frame centrifugal term combine to permit a shared angular speed.

For a small mass ratio $q = m_2/(m_1 + m_2)$, the Hill scale $a(q/3)^{1/3}$ estimates the distance from the smaller body to L1 or L2. L4 and L5 are linearly stable only for sufficiently unequal masses, approximately $q < 0.03852$. Station-keeping may still be needed in a real many-body system.

8. What an orbit can tell you

A measured period and physical orbital size give a mass. An angular size also needs a distance. A radial-velocity curve gives projected speeds, so inclination matters. Before substituting, ask: Is the orbit circular? Is the companion mass negligible? Is the quoted radius a separation, altitude or semimajor axis? Are the units physical or astronomical?

> ***A reliable method.*** Draw the two bodies and the focus. Mark the known radius and velocity direction. Choose energy, angular momentum or the period–mass relation according to the question. Check the circular limit $e \to 0$, the equal-mass barycentre, and the dimensions of the final expression.

9. Lecture + Problems: predict, calculate, check

Fifteen original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — A different planet. A spherical planet has twice Earth's mass and three times its radius. Find its surface gravity and escape speed as multiples of Earth's. Does a larger radius always mean stronger surface gravity?

Worked solution →

[1] Problem 2 — One Earth radius up. A satellite circles at altitude $R_\oplus$ above Earth. Take $g_0 = 9.81\ \mathrm{m\,s^{-2}}$ and $R_\oplus = 6371$ km. Find the local gravity, orbital speed and period.

Worked solution →

[2] Problem 3 — A comet's two speeds. A comet orbits a $1M_\odot$ star with $a = 4$ AU and $e = 0.75$. Find its period, periapsis and apoapsis distances, and its speeds there.

Worked solution →

[2] Problem 4 — The hidden wobble. A circular binary has masses $2M_\odot$ and $1M_\odot$, separated by 3 AU. Find the period, each star's barycentric radius and each orbital speed.

Worked solution →

[2] Problem 5 — Weighing an eclipsing binary. An edge-on circular double-lined binary has period 10 days and radial-velocity semiamplitudes 60 and 120 km/s. Find the mass ratio and both masses. Neglect eccentricity.

Worked solution →

[1] Problem 6 — At the semimajor axis. Use vis-viva to show that $v = \sqrt{GM/a}$ when $r = a$. Does this depend on eccentricity? Explain why it does not mean all such orbits are circular.

Worked solution →

[3] Problem 7 — An ideal Earth–Mars transfer. Treat both planetary orbits as circular and coplanar with radii 1 and 1.524 AU. Find the Hohmann transfer's semimajor axis, travel time and two heliocentric tangential speed increments. Ignore the planets' gravity wells.

Worked solution →

[2] Problem 8 — A sideways boost. At radius r_0 on a circular orbit, a tangential impulse raises the speed to $1.2v_c$. Find the new specific energy, semimajor axis, eccentricity and apoapsis in terms of the original quantities.

Worked solution →

[1] Problem 9 — Can we separate the stars? A binary with total mass $2M_\odot$ and separation 2 AU lies 100 pc away. Find its maximum angular separation in arcseconds and its period.

Worked solution →

[2] Problem 10 — A moon too close. A primary of density $1.5\ \mathrm{g\,cm^{-3}}$ has a satellite of density $3.0\ \mathrm{g\,cm^{-3}}$. Estimate the fluid Roche limit in primary radii. Why is a rigid-body estimate smaller?

Worked solution →

[2] Problem 11 — The Earth–Sun neighbourhood. For $q = 3.0 \times 10^{-6}$ and $a = 1$ AU, estimate the L1 distance using the Hill scale. Convert to kilometres. Is the equilateral-point stability criterion satisfied?

Worked solution →

[2] Problem 12 — Density from a low orbit. A satellite skims a spherical planet with period 90 minutes. Neglect atmosphere and rotation. Show that the planet's mean density is $3\pi/(GT^2)$ and evaluate it.

Worked solution →

[3] Problem 13 — Moon versus Sun in the tides. Use $M_M = 7.342 \times 10^{22}$ kg, $d_M = 3.844 \times 10^8$ m, $M_\odot = 1.98847 \times 10^{30}$ kg and $d_\odot = 1$ AU. Compare their near–far tidal accelerations across Earth's radius 6371 km. Why can the Moon dominate tides while the Sun dominates the gravitational force?

Worked solution →

[3] Problem 14 — Borrowing a planet's motion. A planet moves at 13 km/s. In its frame a probe approaches at 10 km/s and an idealized gravity assist reverses that relative velocity. If the initial heliocentric velocity is $13 - 10$ km/s along the planet's motion, find the final heliocentric velocity and speed gain. Where does the energy come from?

Worked solution →

[3] Problem 15 — The inclination degeneracy. A $1 M_\odot$ star has a single-lined spectroscopic companion with $P = 20$ days, $K_1 = 40$ km/s and $e = 0.2$. The mass function is $f = PK_1^3(1-e^2)^{3/2}/(2\pi G) = m_2^3 \sin^3 i/(m_1+m_2)^2$. Find $f/M_\odot$, the minimum companion mass, and the mass for $i = 30°$. A numerical root is acceptable.

Worked solution →

Further practice and provenance

The 15 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from the papers below. The earlier club lecture compiled these official past-paper identifiers for further practice; only identifiers are reproduced, not copyrighted question statements or official solutions. Round and question labels retain the earlier compilation's notation and should be checked against the official paper.

- IOAA 2007 · TH · 1.6
- IOAA 2007 · TH · 1.7
- IOAA 2009 · TH · P3
- IOAA 2009 · TH · P6
- IOAA 2009 · TH · P12
- IOAA 2009 · TH · P14
- IOAA 2010 · TH · S6
- IOAA 2011 · TH · S1
- IOAA 2011 · TH · S3
- IOAA 2014 · TH · P1
- IOAA 2014 · TH · L3
- IOAA 2015 · TH · S1
- IOAA 2015 · TH · S5
- IOAA 2015 · TH · S13
- IOAA 2016 · TH · T9
- IOAA 2019 · TH · 9

- IOAA 2021 · TH · Q3
- IOAA 2021 · TH · Q9
- IOAA 2021 · TH · Q11
- IOAA 2021 · TH · Q12
- IOAA 2022 · TH · 8
- IOAA 2022 · TH · 12
- IOAA 2023 · TH · Q12
- IOAA 2024 · TH · T3

> ***Sources and scope.*** Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified.